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Reinforcement Learning

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Reinforcement Learning Environments and RLVR

Introduction

Reinforcement Learning (RL) is a machine learning technique where an agent learns by interacting with an environment and receiving rewards or penalties. Modern AI systems now use RL for reasoning, decision-making, and tool usage. Recently, a new concept called Reinforcement Learning from Verifiable Rewards (RLVR) has become important because it uses clear and deterministic rewards instead of human preferences.

The Unsloth and NVIDIA blog “Reinforcement Learning Environments and How to Build Them” explains how RL environments help in training modern AI agents effectively.

Supervised Fine-Tuning (SFT) vs Reinforcement Learning (RL)

Supervised Fine-Tuning (SFT)

SFT trains a model using input-output examples. It is useful for:

- Teaching response formats
- Improving chat behavior
- Learning simple tasks

However, SFT has limitations because models only imitate examples and may fail on unseen situations.

Reinforcement Learning (RL)

RL allows the model to learn by trial and error. The model receives rewards based on whether its actions are correct. RL is more suitable for:

- Reasoning tasks
- Mathematics
- Coding
- Multi-step decision-making

Usually, companies use both methods together: SFT for initial learning and RL for improving reasoning abilities.

What is an RL Environment?

An RL environment is the system in which an AI agent interacts. It includes:

1. The task to perform
2. Available actions
3. Current state of the environment
4. Reward mechanism

The environment checks whether the agent's output is correct and gives rewards accordingly.

Important Terms

- **Rollout:** The process of interacting with the environment.
- **Trajectory:** The sequence of states, actions, and rewards during interaction.

RLVR and GRPO

Traditional RL methods like PPO required large computational resources. A newer method called Group Relative Policy Optimization (GRPO) is more efficient.

Features of GRPO

- Uses deterministic rewards
- Removes the need for separate reward models
- Works well for reasoning tasks

RLVR focuses on verifiable rewards such as:

- Passing unit tests
- Correct tool usage
- Solving puzzles correctly

This makes training more stable and reliable.

Example: Sudoku RL Environment

One simple example discussed in the blog is solving Sudoku puzzles using RL.

How It Works

- The AI agent receives an incomplete Sudoku grid.
- The model generates a completed puzzle.
- A verifier checks whether the solution is correct.

Reward System

- Correct solution → Reward = 1
- Incorrect solution → Reward = 0

Why Sudoku is a Good RL Environment

- Easy to verify solutions
- Binary rewards are simple
- Millions of puzzles can be generated automatically

- No human labeling is needed

This example shows how RLVR can train models efficiently using deterministic feedback.

NVIDIA NeMo Gym

NVIDIA developed an open-source framework called **NeMo Gym** for building scalable RL environments.

Main Components

1. **Agent Server:** Controls interaction with the model
2. **Resource Server:** Provides tools and manages sessions
3. **Model Interface:** Connects the environment with the AI model

NeMo Gym supports large-scale RL training and works with frameworks like Unsloth and Hugging Face TRL.

Real-World Applications

RL environments are now used in:

- AI assistants
- Scientific research agents
- Coding agents
- Tool-calling systems

NVIDIA used RL environments while developing the Nemotron model family.

Conclusion

RL environments are becoming an important part of modern AI systems. RLVR improves AI training by using verifiable rewards instead of human feedback. Methods like GRPO make RL more efficient and scalable. Frameworks such as NVIDIA NeMo Gym help developers build powerful AI agents capable of reasoning and autonomous decision-making.

References

1. Han, D. et al. (2026). *Reinforcement Learning Environments and How to Build Them*. Unsloth & NVIDIA Blog.
2. Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction*. MIT Press.